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In [1]: import numpy as np
import matplotlib.pyplot as plt
from IPython import display
import warnings
warnings.filterwarnings("ignore", category=DeprecationWarning)

try:
    plt.style.use('seaborn-v0_8-white')
except:
    plt.style.use('default')

# Odczytywanie i przetwarzanie danych (Wklejony przykładowy tekst)
data = """
Artificial intelligence (AI) is intelligence-perceiving, synthesizing, and infer
"""

chars = list(set(data))
data_size, X_size = len(data), len(chars)
char_to_ix = { ch:i for i,ch in enumerate(chars) }
ix_to_char = { i:ch for i,ch in enumerate(chars) }

# State i hiperparametry (Wariant 10: H_size=50, T_steps=50)
H_size = 50
T_steps = 50

learning_rate = 1e-1
weight_sd = 0.1
z_size = H_size + X_size

# Funkcje pomocnicze
def sigmoid(x):
    return 1 / (1 + np.exp(-x))

def dsigmoid(y):
    return y * (1 - y)

def dtanh(x):
    return 1 - x * x

# Inicjalizacja wag
Wf = np.random.randn(H_size, z_size) * weight_sd + 0.5
Wi = np.random.randn(H_size, z_size) * weight_sd + 0.5
Wc = np.random.randn(H_size, z_size) * weight_sd
Wo = np.random.randn(H_size, z_size) * weight_sd + 0.5
Wv = np.random.randn(X_size, H_size) * weight_sd
bf = np.zeros((H_size, 1))
bi = np.zeros((H_size, 1))
bc = np.zeros((H_size, 1))
bo = np.zeros((H_size, 1))
bv = np.zeros((X_size, 1))

parameters = {'Wf': Wf, 'Wi': Wi, 'Wc': Wc, 'Wo': Wo, 'Wv': Wv,
              'bf': bf, 'bi': bi, 'bc': bc, 'bo': bo, 'bv': bv}

smooth_grads = {param: np.zeros_like(parameters[param]) for param in parameters}

# Funkcja Forward Pass

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def forward_pass(inputs, targets, h_prev, c_prev):
    xs, hs, cs, zs, os, ys, ps, fs, is_, cs_bar, os_gate = {}, {}, {}, {}, {}, {}
    hs[-1] = np.copy(h_prev)
    cs[-1] = np.copy(c_prev)
    loss = 0

    for t in range(len(inputs)):
        xs[t] = np.zeros((X_size, 1))
        xs[t][inputs[t]] = 1

        zs[t] = np.vstack((hs[t-1], xs[t]))

        fs[t] = sigmoid(np.dot(Wf, zs[t]) + bf)
        is_[t] = sigmoid(np.dot(Wi, zs[t]) + bi)
        cs_bar[t] = np.tanh(np.dot(Wc, zs[t]) + bc)
        os_gate[t] = sigmoid(np.dot(Wo, zs[t]) + bo)

        cs[t] = fs[t] * cs[-1] + is_[t] * cs_bar[t]
        hs[t] = os_gate[t] * np.tanh(cs[t])

        os[t] = np.dot(Wv, hs[t]) + bv
        ys[t] = np.exp(os[t]) / np.sum(np.exp(os[t]))
        loss += -np.log(ys[t][targets[t], 0])

    return loss, fs, is_, cs_bar, os_gate, cs, hs, xs, ys

# Funkcja Backward Pass
def backward_pass(targets, fs, is_, cs_bar, os_gate, cs, hs, xs, ys):
    dWf, dWi, dWc, dWo, dWv = np.zeros_like(Wf), np.zeros_like(Wi), np.zeros_like(Wc), np.zeros_like(Wo), np.zeros_like(Wv)
    dbf, dbi, dbc, dbo, dbv = np.zeros_like(bf), np.zeros_like(bi), np.zeros_like(bc), np.zeros_like(bo), np.zeros_like(bv)

    dh_next = np.zeros((H_size, 1))
    dc_next = np.zeros((H_size, 1))

    grads = {'Wf': dWf, 'Wi': dWi, 'Wc': dWc, 'Wo': dWo, 'Wv': dWv,
             'bf': dbf, 'bi': dbi, 'bc': dbc, 'bo': dbo, 'bv': dbv}

    for t in reversed(range(len(targets))):
        dy = np.copy(ys[t])
        dy[targets[t]] -= 1

        dWv += np.dot(dy, hs[t].T)
        dbv += dy

        dh = np.dot(Wv.T, dy) + dh_next

        dos_gate = dh * np.tanh(cs[t])
        dos_gate = dsigmoid(os_gate[t]) * dos_gate

        dc = dh * os_gate[t] * dtanh(np.tanh(cs[t])) + dc_next

        dfs = dc * cs[t-1]
        dfs = dsigmoid(fs[t]) * dfs

        dis = dc * cs_bar[t]
        dis = dsigmoid(is_[t]) * dis

        dcs_bar = dc * is_[t]
        dcs_bar = dtanh(cs_bar[t]) * dcs_bar

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zs = np.vstack((hs[t-1], xs[t]))

dWf += np.dot(dfs, zs.T)
dbf += dfs
dWi += np.dot(dis, zs.T)
dbi += dis
dWc += np.dot(dcs_bar, zs.T)
dbc += dcs_bar
dWo += np.dot(dos_gate, zs.T)
dbo += dos_gate

dh_next = np.dot(Wf[:, :H_size].T, dfs) + \
            np.dot(Wi[:, :H_size].T, dis) + \
            np.dot(Wc[:, :H_size].T, dcs_bar) + \
            np.dot(Wo[:, :H_size].T, dos_gate)

dc_next = dc * fs[t]

for dparam in grads.values():
    np.clip(dparam, -5, 5, out=dparam)

return grads, hs[-1], cs[-1]

# Funkcja aktualizacji wag
def update_params(grads, params, smooth_grads):
    for param in grads:
        g = grads[param]
        smooth_grads[param] += g * g
        params[param] -= learning_rate * g / (np.sqrt(smooth_grads[param]) + 1e-

# Generowanie tekstu
def sample(h_prev, c_prev, seed_ix, n):
    x = np.zeros((X_size, 1))
    x[seed_ix] = 1
    ixes = []

    h = h_prev
    c = c_prev

    for t in range(n):
        z = np.vstack((h, x))

        f = sigmoid(np.dot(Wf, z) + bf)
        i = sigmoid(np.dot(Wi, z) + bi)
        c_bar = np.tanh(np.dot(Wc, z) + bc)
        o = sigmoid(np.dot(Wo, z) + bo)

        c = f * c + i * c_bar
        h = o * np.tanh(c)

        out = np.dot(Wv, h) + bv
        p = np.exp(out) / np.sum(np.exp(out))

        ix = np.random.choice(range(X_size), p=p.ravel())

        x = np.zeros((X_size, 1))
        x[ix] = 1
        ixes.append(ix)

    return ixes, h, c

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# Główna pętla treningowa
def training_loop():
    global Wf, Wi, Wc, Wo, Wv, bf, bi, bc, bo, bv

    h_prev = np.zeros((H_size, 1))
    c_prev = np.zeros((H_size, 1))

    n, p = 0, 0
    moving_avg_loss = -np.log(1.0/X_size) * T_steps

    while n <= 10000:
        if p + T_steps + 1 >= data_size or n == 0:
            h_prev = np.zeros((H_size, 1))
            c_prev = np.zeros((H_size, 1))
            p = 0

        inputs = [char_to_ix[ch] for ch in data[p:p + T_steps]]
        targets = [char_to_ix[ch] for ch in data[p + 1:p + T_steps + 1]]

        loss, fs, is_, cs_bar, os_gate, cs, hs, xs, ys = forward_pass(inputs, ta
        moving_avg_loss = moving_avg_loss * 0.999 + loss * 0.001

        grads, h_prev, c_prev = backward_pass(targets, fs, is_, cs_bar, os_gate,
        update_params(grads, parameters, smooth_grads)

        if n % 500 == 0:
            seed_ix = inputs[0] if inputs else char_to_ix[chars[0]]
            sample_ix, _, _ = sample(h_prev, c_prev, seed_ix, 200)
            txt = ''.join(ix_to_char[ix] for ix in sample_ix)

            print(f'Iteracja: {n}, Strata: {moving_avg_loss:8.4f}')
            print('--- GENEROWANY TEKST ---')
            print(txt)
            print('-----')

        p += T_steps
        n += 1

    print(f'Trening zakończony po {n-1} iteracjach. Ostateczna strata: {moving_a

# Uruchomienie pętli treningowej (GŁÓWNY KROK)
print("Rozpoczynanie treningu dla Wariantu 10 (H_size=50, T_steps=50, dane: Wkle
training_loop()

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Rozpoczynanie treningu dla Wariantu 10 (H_size=50, T_steps=50, dane: Wklejony tekst)...

Iteracja: 0, Strata: 186.8835

--- GENEROWANY TEKST ---

ClEmwCoGv cp(-a lc,xI.lri(fgzIniIk
Ir i(rhg-D(,jDrzvtA-IrvDnStnEccvzulll,idi.(nmirior tysAgiCIv cIdu())EGklGnTblin,y
o,rhAtr)Ihn)IrnIkfk(cnacxmnIcetzto)))gi sri e-i)ecGxcj.iET-ttl Eysipclt-Ar(iwigi
hck

Iteracja: 500, Strata: 173.3460

--- GENEROWANY TEKST ---

m'ttImilii gtliiifticttibgtfgitsnnregtgtunsgelIett crnet(itt ian iincty ntttcgcil
cm ilttyleAr nu uttiytktgiatAtigtgt inc i ecao aIu ursctiieitciledh tiietthgiI
tgrscstt lasgxs.aicnt-icto)sc-moitrtfa

Iteracja: 1000, Strata: 162.6485

--- GENEROWANY TEKST ---

mdrrdgi e fhi uttaon flrredhylheemberngn sfe.phas hespmlsc locsnecnof d me atol
it lpbabvilulydahn dssle ame b nbar l elben nhaieneeul,arrantsovspcnegictddynoi
sdsIgs ayaebaaliargblndndeuloosenhnnnn

Iteracja: 1500, Strata: 155.7347

--- GENEROWANY TEKST ---

snIlEIetr.-,griIitEh-triGnsitp.TtEE.tt-.Giuit..r lnq.i.tleGr rrrhirt .rltiItCmirt
rttsrThuiECih. tqEE,.qinrgIrliEIhrr.rristrrrerriuiEEIitrft,i.cti'tiE.. .tEr tn.T
,Tehm.rphtrtvEaviTt.Gn.r.niT,h Ta's'G

Iteracja: 2000, Strata: 151.1680

--- GENEROWANY TEKST ---

Inc aseesaeeee eee-m.rr.hT ee
nreeeio,keseuett .teGriKh.rraieeine teei)ia,.u,,ut e.T.Tafv-etdnsTtad set v rgts
i f rprtc eaelp rrheoA tAar taua rarIreeano snnroe peo uhreeeeeetoeeeekeeeeee,e
rete

Iteracja: 2500, Strata: 147.7158

--- GENEROWANY TEKST ---

ned ,ren lnosnt inr r ttitomhnnysaoe i alrnsiltvts lAl iitntztgi(ieet oseti,d
uarpmmt g Afc r f lg lamvte et ml p eu r ueitettee s eiaai et tl teea
epiytgtgrlt nanlvl npnnlrtrlrpevlprtt

Iteracja: 3000, Strata: 145.3468

--- GENEROWANY TEKST ---

oesn mi no sig lntorsne lncsit ats l aiositetlsnat ieaa tneosdvact nbl(atrae n
leci ymsvra eli m noa yeaspl c aavl ilsy estm ate tsurrranl ie tlg(itlnodov
amhltm eetpe aeaaoin nnrAd na -tntonoc

Iteracja: 3500, Strata: 143.9175

--- GENEROWANY TEKST ---

nryieltsytori ijfpyaastbtr gei, t -st tt,tgaeru ab ta etjat ktrteaaiuareessebo
llni aebbt(misehhlh(u.hht.ihe gif(lauge-hhehr riAt bhr htuiiAhhrhhhhhrboenn retra
m .emesemmeoeueho attee ot teeeme oit

Iteracja: 4000, Strata: 142.7248

--- GENEROWANY TEKST ---

eas iAoieypeae m ,e, eaoa n annn u)sieiIluppho ityhjihhhhhhhhhhhhh.hhhhhhh.oqi
hhhheahhrhhhhhmaai aoeraeqaerrherotqaitqaasAataAaAttaAranketaetGtAeA pp eetetye
eonnennnnneaneoeebd uta ulaan nnn ,t

Iteracja: 4500, Strata: 141.6329

--- GENEROWANY TEKST ---

cTsnbhtcm (telta att atooa-eyp)atlaat alo)GaphsaasxxxntlvaxagrgTTajTggTTigTATjax
ianTriaTiAaal alaaaaAipaaJyap ylnarAaepiyAr nrtenrnnfaenapniaaannai nrarnnnnnraa
rn n nac rrrrrr accancppnra,rnrassnnr a

Iteracja: 5000, Strata: 140.7427

--- GENEROWANY TEKST ---

ingenArhaler(tyeg,ioete a, eeb, obaasgaaenvsfta, drslmrrmtlnlnlnnlihlennla rcthaan
lbnhhahxrrmrxxnclynTTglggpgTsgrrtTthmau rasaramtaaaiuameDapanaAAegaeiaATar aaa.aa
aimpaxpAAaa aanuxuientnatnnnetnantnnnnn

Iteracja: 5500, Strata: 140.2136

--- GENEROWANY TEKST ---

cigpczm leslaaintapno. te,rttonaa abauahedtmtnannnlrrtxtAnnAvAAAvflnnlllxxalnavn
nDlGnnanangcx lnlllgytxgilTDT(vtecTgccxGg(AeiaTlebn. aaaxxt.snaanansnn na a nra
aonaenan tcnlealeaeaana neeanaieseaea

Iteracja: 6000, Strata: 139.6750

--- GENEROWANY TEKST ---

Gerrg catey mcalntaatg te,e,wyt ,,t,eaiaah aaeltaalnacyaaacgihtaeuhvjihmsshsoh
arelhohehnha helhhherhrnhhhhehhcirhrrmhihhhlhnneuhehrrrelrrhyhheherkaylhheuhel h
Gusetmhasroenhlraehusectnhhlvhrhhoheh

Iteracja: 6500, Strata: 140.1243

--- GENEROWANY TEKST ---

dn l l urnloolosonaa T feageigaosdTgallva lags,rr-aggbc geA gtne ttoirttmeenantp
en na, nataagro ,a,calettaanyanaranaaaa cag)g eggi aggh honaaamla ataoeaagirsoaeg
eop) eean oetaugnae)c aonoucae gaataa

Iteracja: 7000, Strata: 140.1505

--- GENEROWANY TEKST ---

dtsnen cnytlczeoaiiplaeaidtotdcctlgcctci teoarna gn,ate ,aotadtea,aba Aaea tazaa
unaa e aaamapaa utla oduoehaeap phe eomiaaaaurvaIeeeietaua nateaogaafduae gh
iaa aeahuaeeh pgoipiaa oaoaaIe apbv

Iteracja: 7500, Strata: 139.8100

--- GENEROWANY TEKST ---

pgorypnignyhfhlmngtttdieae (gml, gogag,iodaacigcettetta,tannteppapr o pnlaheah nh
uheIhemseahahso mo hl ithlut ihaulainnnmunssenr a(hst,tnt la llpaucalllvcestteh
lnltnn tnt tntllrnrtrrutlittlttcarlhsgn

Iteracja: 8000, Strata: 139.4895

--- GENEROWANY TEKST ---

sf nvtani jAilpigliileytrmaggoe,olggdgh nouloohuhemIaln tvaaatakr lttgl nkftgnAta
llrnxlsltttra,totml ansanlansyanranagnlgrscansnaagAnnnons(arnaannnprnsnsfaanahsax
nai(racpaTt Caapgaarn(aevAp)Aat (a a

Iteracja: 8500, Strata: 139.5825

--- GENEROWANY TEKST ---

nrmellcvilnisjdr (taazubdadegtAeodbt rdldstnareoaatat olea ejua ea ter a,otIrannt
dmve ,nua gcedavhu tamaaojnhhioiaah ahaenkvagchhmrhge hatabavtrhhmx ki o.(lygaa
rcathrhxchhauharh l hhhshembrahahshnh

Iteracja: 9000, Strata: 140.0914

--- GENEROWANY TEKST ---

oastk ovalsi.(aaodtpteooyaorh dhptg ,aasdgraaugnae fsaieeiaheoege hh eingged
ateaei heaaetsetaama elacealigcaaga ain uaigocve gaagtegtt ae e)gcaggae agaa)a
hatttaaguueantegouccacanyc iaauaavage

Iteracja: 9500, Strata: 140.2175

--- GENEROWANY TEKST ---

sionidvaricttiai olsglaaatttf aa toae-aaiehheh saehiuetttu lmeohjmssbupneaanawlm
,tbasuotliqvoa bi ostIua aisdfa ,bagaea rtsdeananilaiin t eabeh as lh hdtaephnlre
eaibIeonhalhnapao.aaennuteuss,ia nfeir

Iteracja: 10000, Strata: 139.9178

--- GENEROWANY TEKST ---

rs tiplptiiofa Aaifdtatctceaito,aatetti maoroo al oaamenataiozlaacebnatuaaaaaaaau
faeaaaag eaeheteeaeae o-aea)zgeuuaaeaca egagateehteiteeiaefigutdht atagceotaaataga
ieoaatyleteseagaargne cgh(aagefeetage

Trening zakończony po 10000 iteracjach. Ostateczna strata: 139.9178